Ultrafast Transient Grating Spectroscopy of GaAs

Dhruv Pisharody Ahmet Kavruk
Angelo Brade
Rheinische Friedrich–Wilhelms–Universität Bonn

Abstract

Transient-grating spectroscopy separates carrier recombination from spatial diffusion by measuring the decay of a spatially periodic carrier population. A Gaussian fit to the supplied spectrum gives a peak near 788.66 nm and an FWHM bandwidth of 10.92 nm, corresponding to a transform-limited duration of about 84 fs. Fits to the raw autocorrelation traces, using the tutorial's fundamental and second-harmonic fringe terms, give (81.5 ± 0.3) fs before and (79.0 ± 0.1) fs after the compressor (repeat-to-repeat standard deviations); the incomplete `vor2` scan is excluded. The local fringe calibration assumes nearly constant delay-scan speed across the fitted peak. The supplied data do not provide enough calibration information to determine hot-carrier cooling time or absolute carrier diffusivity and mobility.

1 Theoretical background

An ultrashort pump pulse creates electron–hole pairs in GaAs. Two pump fields with equal intensity I_0 and wave vectors $\mathbf{k}_1, \mathbf{k}_2$ produce the intensity pattern

$$I(\mathbf{r}) = |E_1 + E_2|^2 \tag{1}$$
$$= |E_1|^2 + |E_2|^2 + E_1 E_2^* + E_1^* E_2 \tag{2}$$
$$= 2I_0 \left[1 + \cos\left((\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r} + \phi\right)\right]. \tag{3}$$

For an angle θ between the beams, $q = |\mathbf{k}_1 - \mathbf{k}_2| = 2k \sin(\theta/2)$ and

$$\Lambda_{\mathrm{TG}} = \frac{2\pi}{q} = \frac{\lambda}{2\sin(\theta/2)}. \tag{4}$$

The two-beam interference and resulting spatial carrier modulation are illustrated in Fig. 1.

Write the carrier density as a uniform part plus its sinusoidal grating component,

$$n(x,t) = N_0(t) + N_q(t)\cos(qx). \tag{5}$$

Substitution into the tutorial's one-dimensional diffusion–recombination equation gives

$$\frac{\partial n(x,t)}{\partial t} = D_a \frac{\partial^2 n(x,t)}{\partial x^2} - \frac{n(x,t)}{\tau_{\mathrm{rec}}}. \tag{6}$$

Using $\partial_t n = \dot{N}_0 + \dot{N}_q \cos(qx)$ and $\partial_x^2 n = -q^2 N_q \cos(qx)$, and equating the coefficients of 1 and $\cos(qx)$, respectively,

$$\dot{N}_0 = -\frac{N_0}{\tau_{\mathrm{rec}}}, \qquad \dot{N}_q = -\left(D_a q^2 + \frac{1}{\tau_{\mathrm{rec}}}\right) N_q. \tag{7}$$

$$\int_{N_0(0)}^{N_0(t)} \frac{dN_0}{N_0} = -\int_0^t \frac{dt'}{\tau_{\mathrm{rec}}}, \qquad \int_{N_q(0)}^{N_q(t)} \frac{dN_q}{N_q} = -\int_0^t \left(D_a q^2 + \frac{1}{\tau_{\mathrm{rec}}}\right) dt'. \tag{8}$$

Evaluation yields

$$N_0(t) = N_0(0)e^{-t/\tau_{\rm rec}}, \tag{9}$$

$$N_q(t) = N_q(0)e^{-(D_a q^2 + 1/\tau_{\rm rec})t}. \tag{10}$$

For signals proportional to these populations, the $q = 0$ transient-reflectivity and finite-q grating traces therefore have the forms

$$S_{\rm pop}(t) = A_{\rm pop}e^{-t/\tau_{\rm rec}} + C_{\rm pop}, \tag{11}$$

$$S_{\rm TG}(t) = A_{\rm TG}e^{-t/\tau_{\rm TG}} + C_{\rm TG}, \tag{12}$$

$$\frac{1}{\tau_{\rm TG}} = \frac{1}{\tau_{\rm rec}} + D_a q^2. \tag{13}$$

In $1/\tau_{\rm TG} = 1/\tau_{\rm rec} + D_a q^2$, the intercept is $1/\tau_{\rm rec}$ and the slope is D_a.

For ambipolar transport, set the excess electron and hole densities equal, $n = h$, and equate their particle fluxes. With the drift–diffusion expressions

$$\mathbf{j}_e = -D_e \nabla n - \mu_e n \mathbf{E}, \qquad \mathbf{j}_h = -D_h \nabla n + \mu_h n \mathbf{E}, \tag{14}$$

the condition $\mathbf{j}_e = \mathbf{j}_h$ gives

$$-(D_e - D_h)\nabla n = (\mu_e + \mu_h) n \mathbf{E}, \tag{15}$$

$$\mathbf{E} = \frac{D_h - D_e}{\mu_e + \mu_h}\frac{\nabla n}{n}, \tag{16}$$

$$\mathbf{j}_e = -\left[D_e + \mu_e \frac{D_h - D_e}{\mu_e + \mu_h}\right]\nabla n \tag{17}$$

$$= -\frac{D_e\mu_h + D_h\mu_e}{\mu_e + \mu_h}\nabla n. \tag{18}$$

Comparing with Fick's law $\mathbf{j} = -D_a \nabla n$ identifies

$$D_a = \frac{D_e\mu_h + D_h\mu_e}{\mu_e + \mu_h}, \tag{19}$$

$$D_i = \frac{k_B T}{e}\mu_i \quad (i = e, h) \tag{20}$$

$$\Longrightarrow \quad D_a = \frac{2\mu_e\mu_h}{\mu_e + \mu_h}\frac{k_B T}{e} = \mu_a \frac{k_B T}{e}, \tag{21}$$

$$\mu_a = \frac{2\mu_e\mu_h}{\mu_e + \mu_h}. \tag{22}$$

The Einstein step assumes non-degenerate carriers at a common temperature. At high carrier density, many-body and degeneracy effects can change D_a [4].

For a Gaussian pulse intensity $I(t) = I_0 e^{-t^2/(2\sigma_t^2)}$, the intensity autocorrelation is

$$G(\tau) = \int_{-\infty}^{\infty} I(t)I(t-\tau)\,dt \tag{23}$$

$$= I_0^2 \int_{-\infty}^{\infty} \exp\left[-\frac{t^2 + (t-\tau)^2}{2\sigma_t^2}\right] dt \tag{24}$$

$$= I_0^2 e^{-\tau^2/(4\sigma_t^2)} \int_{-\infty}^{\infty} e^{-(t-\tau/2)^2/\sigma_t^2}\,dt \tag{25}$$

$$= I_0^2 \sqrt{\pi}\sigma_t e^{-\tau^2/(4\sigma_t^2)}. \tag{26}$$

Here $t^2 + (t-\tau)^2 = 2(t-\tau/2)^2 + \tau^2/2$ and the substitution $u = (t-\tau/2)/\sigma_t$ reduces the integral to $\sigma_t \int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}\sigma_t$. The intensity half-maximum satisfies $e^{-t^2/(2\sigma_t^2)} = 1/2$, so

$\Delta t_{\text{pulse}} = 2\sqrt{2\ln 2}\,\sigma_t$. The autocorrelation half-maximum satisfies $e^{-\tau^2/(4\sigma_t^2)} = 1/2$, so $\Delta t_{\text{AC}} = 4\sqrt{\ln 2}\,\sigma_t = \sqrt{2}\,\Delta t_{\text{pulse}}$.

The Gaussian time–bandwidth product follows by Fourier transforming the field envelope $a(t) = \sqrt{I_0}e^{-t^2/(4\sigma_t^2)}$:

$$\widetilde{a}(\nu) = \int_{-\infty}^{\infty} a(t)e^{-i2\pi\nu t}\,dt \tag{27}$$

$$= \sqrt{I_0}\,e^{-4\pi^2\sigma_t^2\nu^2}\int_{-\infty}^{\infty}\exp\left[-\frac{(t+i4\pi\sigma_t^2\nu)^2}{4\sigma_t^2}\right]dt \tag{28}$$

$$= 2\sqrt{\pi}\sigma_t\sqrt{I_0}\,e^{-4\pi^2\sigma_t^2\nu^2}, \tag{29}$$

$$|\widetilde{a}(\nu)|^2 = 4\pi\sigma_t^2 I_0 e^{-8\pi^2\sigma_t^2\nu^2}. \tag{30}$$

Setting the spectral intensity to half its peak gives $|\nu| = \sqrt{\ln 2}/(2\sqrt{2}\pi\sigma_t)$, hence $\Delta\nu = \sqrt{\ln 2}/(\sqrt{2}\pi\sigma_t)$ and

$$\Delta t_{\text{pulse}}\Delta\nu = 2\sqrt{2\ln 2}\sigma_t\frac{\sqrt{\ln 2}}{\sqrt{2}\pi\sigma_t} = \frac{2\ln 2}{\pi} \simeq 0.441. \tag{31}$$

From $\nu = c/\lambda$, differentiation gives $d\nu/d\lambda = -c/\lambda^2$; for a narrow spectrum centered at λ_0, $\Delta\nu \simeq (c/\lambda_0^2)\Delta\lambda$.

The optical pulse duration can be measured with a second-order autocorrelator based on second-harmonic generation (SHG). The measured intensity is a function of the relative delay between two pulse replicas. For a Gaussian pulse, the intensity autocorrelation FWHM is $\sqrt{2}$ times the pulse-intensity FWHM. The transform-limited Gaussian time-bandwidth product is $\Delta t\Delta\nu \simeq 0.441$; a larger product can indicate spectral phase/chirp, or measurement and calibration errors. A spectrometer measures wavelength, with $\Delta\nu \simeq c\Delta\lambda/\lambda_0^2$ for a narrow spectrum. This conversion and the time-bandwidth comparison assume that the fitted spectral profile is representative of the pulse spectrum and approximately Gaussian.

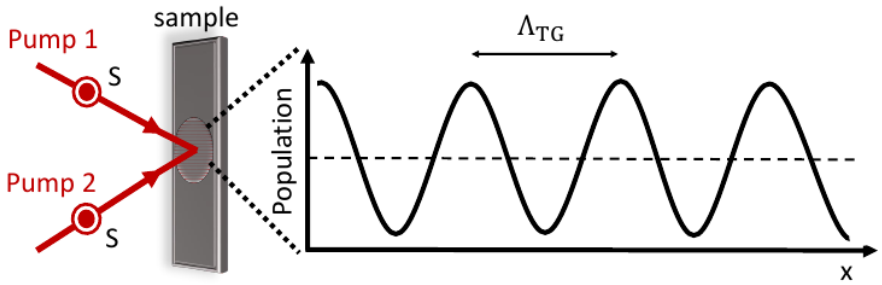


Figure 1: Two crossed pump beams create a sinusoidal excitation pattern in the sample. The grating period Λ_{TG} sets the spatial wave number $q = 2\pi/\Lambda_{\text{TG}}$ used in the diffusion analysis. Reproduced from Fig. 4.5 of the course tutorial.

2 Setup

The tutorial describes a Kerr-lens mode-locked Ti:sapphire femtosecond laser. Its intensity-dependent refractive index produces self-focusing that favors pulsed operation; a prism compressor compensates dispersion and reduces pulse chirp. For autocorrelation, a Michelson interferometer creates two pulse replicas with adjustable delay and a BBO crystal generates a second-harmonic signal proportional to their temporal overlap. For the sample measurement, the beam is split into pump and probe paths; two pump beams cross on the GaAs sample and make the excitation grating. Two probe beams arranged in BOXCAR geometry provide

a diffracted signal and a local oscillator. Their interference gives a heterodyne term proportional to the diffracted field, improving sensitivity over direct intensity detection. The tutorial describes phase modulation of the local oscillator with a rotating coverslip and pump modulation with a chopper; a lock-in amplifier extracts the component at the chopper frequency. The overall optical layout is shown in Fig. 2; the individual Michelson autocorrelator and BOXCAR detection principle are shown in Figs. 3 and 4. These are the tutorial's apparatus and procedure; the provided data do not document which alignment settings or operating parameters were used in this particular run. The files contain repeated delay-signal samples, a wavelength spectrum, and auxiliary traces named `vor*` and `nach*`. Incidence angles, calibrated grating period, sample orientation, pump fluence, sample temperature, and delay-stage calibration were not supplied.

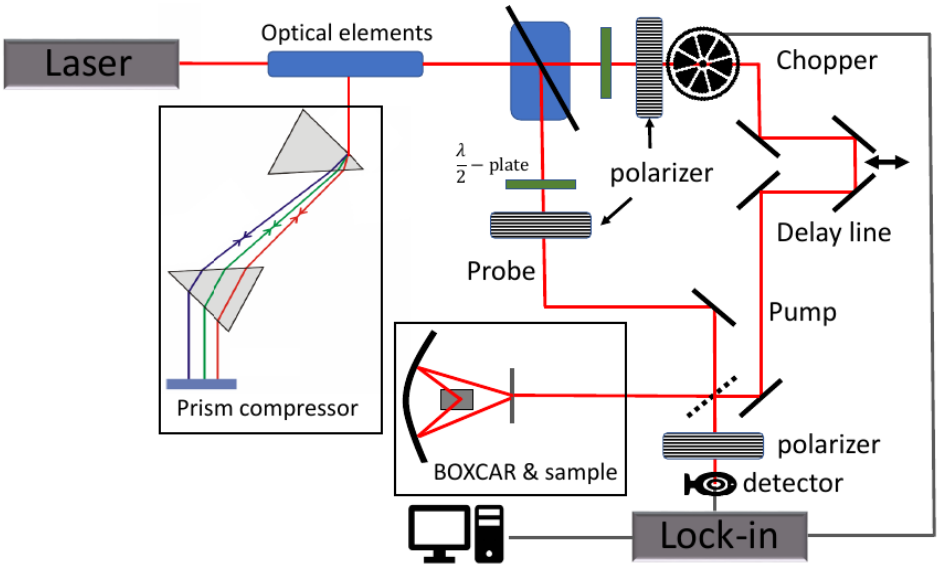


Figure 2: Tutorial schematic of the transient-reflectivity and transient-grating apparatus, including prism compressor, pump and probe paths, delay line, chopper, BOXCAR sample geometry, detector, and lock-in amplifier. Reproduced from Fig. 4.7 of the course tutorial.

3 Measurement and analysis

The analysis script `analysis/analyze.py` averages repeated signal columns, estimates pointwise standard errors for plotting, fits the spectrum to a Gaussian plus constant background, and analyzes the autocorrelation traces. TG traces are fitted to a single exponential plus offset over the nominal interval 0.5–15 dataset units. Repeated columns are treated as repeat measurements. The suffixes are provisionally interpreted as grating periods in micrometres only to make the exploratory q^2 plot; this interpretation is unverified. The TG delay unit is also unknown, so the fitted lifetimes and rate slope are reported only in dataset coordinates. The autocorrelation files are oscilloscope traces rather than calibrated optical-delay scans; their conversion uses the local optical fringe period and assumes the scan speed is approximately constant over the fitted peak. The measured second-order autocorrelation is given in the tutorial by Eq. (4.2). With

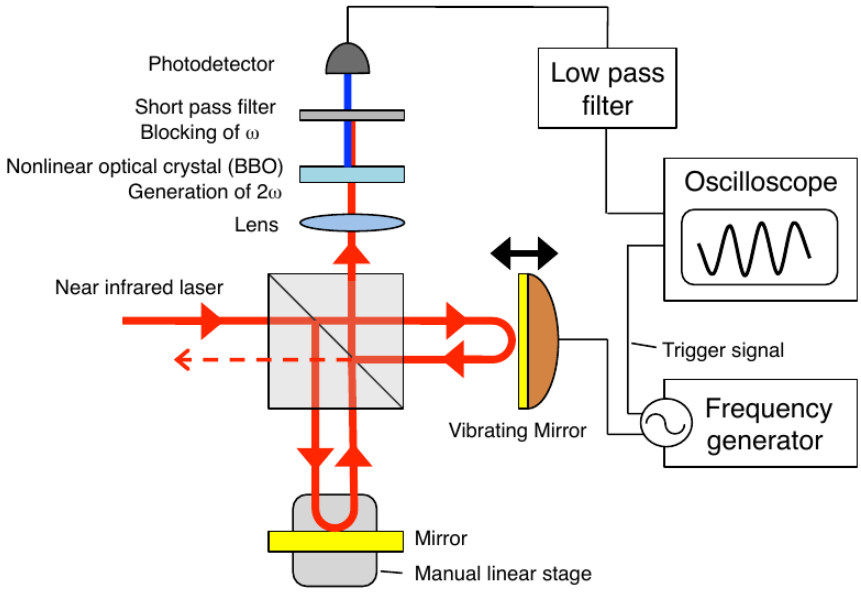


Figure 3: Michelson autocorrelator schematic: the vibrating mirror varies the relative delay between pulse replicas, and a BBO crystal generates the detected second harmonic. Reproduced from Fig. 4.1 of the course tutorial.

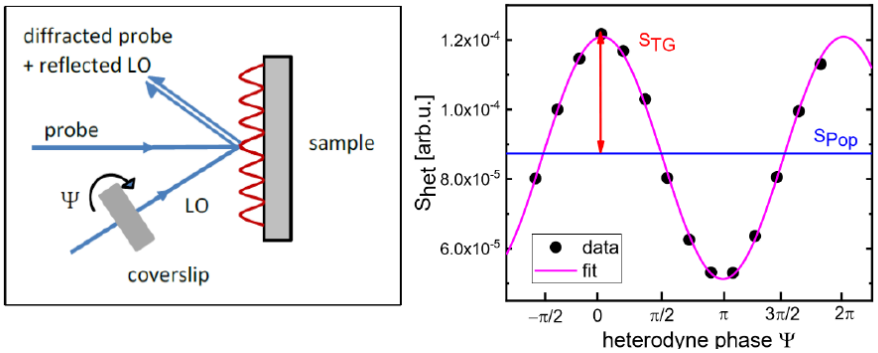


Figure 4: BOXCAR heterodyne detection of the diffracted probe and reflected local oscillator, with the measured signal illustrated as a function of heterodyne phase. Reproduced from Fig. 4.6 of the course tutorial.

$I(t) = |E(t)|^2$, its full expression is

$$\begin{aligned} I_{\mathrm{AC}}(\tau) = & \int_{-\infty}^{\infty} \left[I^2(t) + I^2(t-\tau)\right] \, \mathrm{d}t \\ & + \int_{-\infty}^{\infty} \left[I(t) + I(t-\tau)\right] \mathrm{Re}\{E(t)E^*(t-\tau)\} \, \mathrm{d}t \\ & + \int_{-\infty}^{\infty} \mathrm{Re}\left\{E^2(t)E^{*2}(t-\tau)\right\} \, \mathrm{d}t \\ & + \int_{-\infty}^{\infty} I(t)I(t-\tau) \, \mathrm{d}t. \end{aligned} \tag{4.2}$$

For explicit evaluation, take a Gaussian field

$$E(t) = E_0 e^{-t^2/(4\sigma^2)} e^{-i\omega_0 t}, \qquad I(t) = |E(t)|^2 = I_0 e^{-t^2/(2\sigma^2)}, \qquad I_0 = |E_0|^2. \tag{32}$$

The Gaussian integral used below is obtained by squaring it and converting to polar coordinates:

$$J = \int_{-\infty}^{\infty} e^{-u^2} \, du, \tag{33}$$

$$J^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(u^2+v^2)} \, du \, dv = \int_0^{2\pi} d\varphi \int_0^{\infty} e^{-r^2} r \, dr = 2\pi \left[-\frac{e^{-r^2}}{2}\right]_0^{\infty} = \pi, \tag{34}$$

$$J = \sqrt{\pi}. \tag{35}$$

The four integrals in Eq. (4.2) then evaluate as follows. The first, delay-independent term is

$$T_0 = \int_{-\infty}^{\infty} \left[I^2(t) + I^2(t-\tau)\right] dt \tag{36}$$

$$= 2I_0^2 \int_{-\infty}^{\infty} e^{-t^2/\sigma^2} \, dt = 2I_0^2 \sqrt{\pi}\sigma. \tag{37}$$

For the fundamental-frequency term, first multiply each intensity by the field overlap. The carrier factor is

$$E(t)E^*(t-\tau) = I_0 e^{-[t^2+(t-\tau)^2]/(4\sigma^2)} e^{-i\omega_0\tau}. \tag{38}$$

The two overlap integrals are equal after completing the square:

$$\int_{-\infty}^{\infty} I(t)E(t)E^*(t-\tau) \, dt = I_0^2 e^{-i\omega_0\tau} \int e^{-[3t^2+(t-\tau)^2]/(4\sigma^2)} \, dt \tag{39}$$

$$= I_0^2 e^{-i\omega_0\tau} e^{-3\tau^2/(16\sigma^2)} \int e^{-(t-\tau/4)^2/\sigma^2} \, dt \tag{40}$$

$$= I_0^2 \sqrt{\pi}\sigma e^{-3\tau^2/(16\sigma^2)} e^{-i\omega_0\tau}, \tag{41}$$

$$\int_{-\infty}^{\infty} I(t-\tau)E(t)E^*(t-\tau) \, dt = I_0^2 \sqrt{\pi}\sigma e^{-3\tau^2/(16\sigma^2)} e^{-i\omega_0\tau}. \tag{42}$$

Taking the real part of their sum gives

$$T_1 = 2I_0^2 \sqrt{\pi}\sigma e^{-3\tau^2/(16\sigma^2)} \cos(\omega_0\tau). \tag{43}$$

For the second-harmonic field-overlap term,

$$E^2(t)E^{*2}(t-\tau) = I_0^2 e^{-[t^2+(t-\tau)^2]/(2\sigma^2)} e^{-i2\omega_0\tau}, \tag{44}$$

$$T_2 = \mathrm{Re} \int_{-\infty}^{\infty} E^2(t)E^{*2}(t-\tau) \, dt \tag{45}$$

$$= I_0^2 e^{-\tau^2/(4\sigma^2)} \cos(2\omega_0\tau) \int e^{-(t-\tau/2)^2/\sigma^2} \, dt \tag{46}$$

$$= I_0^2 \sqrt{\pi}\sigma e^{-\tau^2/(4\sigma^2)} \cos(2\omega_0\tau). \tag{47}$$

The final intensity-overlap term has the same Gaussian integral without the cosine:

$$T_3 = \int_{-\infty}^{\infty} I(t)I(t-\tau)\,dt \tag{48}$$

$$= I_0^2 e^{-\tau^2/(4\sigma^2)} \int e^{-(t-\tau/2)^2/\sigma^2}\,dt \tag{49}$$

$$= I_0^2 \sqrt{\pi}\sigma e^{-\tau^2/(4\sigma^2)}. \tag{50}$$

Combining these evaluated terms gives the Gaussian-pulse form of Eq. (4.2):

$$I_{\mathrm{AC}}(\tau) = 2I_0^2\sqrt{\pi}\sigma + 2I_0^2\sqrt{\pi}\sigma e^{-3\tau^2/(16\sigma^2)}\cos(\omega_0\tau) + I_0^2\sqrt{\pi}\sigma e^{-\tau^2/(4\sigma^2)}[1+\cos(2\omega_0\tau)]. \tag{51}$$

Thus the integrals explicitly produce fringes at ω_0 and $2\omega_0$. For the measured trace, the fit uses a common Gaussian envelope for the slowly varying terms as a phenomenological approximation to these overlap functions; it fits

$$S(x) = S_0 + e^{-(x-x_0)^2/(2\sigma_x^2)}\left[A_0 + A_1\cos(2\pi f(x-x_0)+\phi_1) + A_2\cos(4\pi f(x-x_0)+\phi_2)\right]. \tag{52}$$

In a scan coordinate x with fringe frequency f, one optical cycle takes $1/f$ on the recorded axis and λ_0/c in delay. Therefore

$$\tau - \tau_0 = \frac{\lambda_0}{c} f(x-x_0), \tag{53}$$

$$\omega_0(\tau-\tau_0) = \frac{2\pi c}{\lambda_0}\frac{\lambda_0}{c} f(x-x_0) = 2\pi f(x-x_0), \tag{54}$$

$$2\omega_0(\tau-\tau_0) = 4\pi f(x-x_0), \tag{55}$$

which gives the two cosine arguments in the fit. The fitted Gaussian intensity envelope is

$$I_{\mathrm{env}}(x) = A_{\mathrm{env}}\exp\left[-\frac{(x-x_0)^2}{2\sigma_x^2}\right], \qquad A_{\mathrm{env}} = A_0 + |A_1| + |A_2|. \tag{56}$$

In the tutorial's field-envelope notation, its square root is

$$E_{\mathrm{env}}(x) = \sqrt{I_{\mathrm{env}}(x)} = \sqrt{A_{\mathrm{env}}}\exp\left[-\frac{(x-x_0)^2}{4\sigma_x^2}\right], \qquad E_{\mathrm{env}}^2(x) = I_{\mathrm{env}}(x). \tag{57}$$

The appendix plot overlays the normalized $E_{\mathrm{env}}(x)$ (the square root of the fitted intensity-envelope) on the raw signal. The fit envelope has standard deviation σ_x in recorded coordinate. The local scan factor is $d\tau/dx = f\lambda_0/c$, so $\sigma_\tau = (f\lambda_0/c)\sigma_x$. Its FWHM and the pulse FWHM are

$$\Delta t_{\mathrm{AC}} = 2\sqrt{2\ln 2}\,\sigma_\tau = 2\sqrt{2\ln 2}\,\sigma_x f\lambda_0/c, \tag{58}$$

$$\Delta t_{\mathrm{pulse}} = \frac{\Delta t_{\mathrm{AC}}}{\sqrt{2}}. \tag{59}$$

This conversion assumes a locally constant delay-scan speed. The fitted traces are shown in Fig. 9. We exclude `vor2`: its recorded scan ends at 2.8 ms, compared with 14 ms for `vor1` and `vor3`, leaving the outer autocorrelation wings incompletely recorded (Fig. 8). Its sampling interval also differs by a factor of five, so its scan calibration is not directly comparable. Including it would bias the before-compressor pulse-width estimate.

The measured spectrum and Gaussian fit are shown in Fig. 5. The spectrum file spans approximately 345 nm to 1041 nm. We fitted all recorded points with

$$S(\lambda) = B + A\exp\left[-\frac{(\lambda-\lambda_0)^2}{2\sigma_\lambda^2}\right]. \tag{60}$$

The peak above baseline is A and the fitted peak level is $B + A$. At half peak height above baseline, $S - B = A/2$, so

$$e^{-(\lambda-\lambda_0)^2/(2\sigma_\lambda^2)} = \frac{1}{2}, \tag{61}$$

$$|\lambda - \lambda_0| = \sqrt{2\ln 2}\,|\sigma_\lambda|, \tag{62}$$

$$\Delta\lambda_{\mathrm{FWHM}} = 2\sqrt{2\ln 2}\,|\sigma_\lambda|. \tag{63}$$

The fit gives a centre of 788.66 nm, peak intensity of 13 302 a u , median measured intensity of 2501 a u , and fitted half-maximum level of 7902 a u ; $\Delta\lambda_{\mathrm{FWHM}} \approx 10.92$ nm. Applying $\Delta\nu \simeq c\Delta\lambda/\lambda_0^2$ gives $\Delta\nu \approx 5.26$ THz. The Gaussian time-bandwidth result derived above gives $\Delta t_{\mathrm{TL}} = 0.441/\Delta\nu \approx 83.8$ fs. The median is calculated from all measured spectrum points; it is plotted as a descriptive reference and is not the fitted background B.

The direct fits to the raw interferometric autocorrelation traces give a mean pulse duration of (81.5 ± 0.3) fs before and (79.0 ± 0.1) fs after the compressor. The uncertainties are sample standard deviations between the two accepted before-compressor scans and the three after-compressor scans; they quantify repeat scatter only, not total measurement uncertainty. The corresponding time-bandwidth products are about 0.43 and 0.42. They are close to the Gaussian transform limit of 0.441, but the apparent compressor-associated change is only about 2.5 fs; without an independent delay-stage calibration and systematic error budget, its significance cannot be established.

4 Results

Quantity	Result from supplied files
Gaussian-fit spectral peak	788.66 nm
Gaussian-fit peak intensity	13 302 a u
Measured median intensity (plotted baseline reference)	2501 a u
Fitted half-maximum level	7902 a u
Gaussian-fit spectral FWHM	10.92 nm
Approximate frequency bandwidth	5.26 THz
Gaussian transform-limited pulse duration	83.8 fs
Measured duration before compressor	(81.5 ± 0.3) fs
Measured duration after compressor	(79.0 ± 0.1) fs
Time-bandwidth product before / after	0.43 / 0.42
Cooling time and electron-phonon coupling	Not determinable from available labeled data
τ_{rec}, D_a, and μ_a	Not reliably determinable

The individual exploratory TG fit lifetimes and the provisional period interpretation are listed in Table 1. The $q = 0$ pump–probe file is treated as transient reflectivity; the other suffixes are provisionally read as periods in micrometres. These fits are retained to show what the data files contain, but their time axis is not calibrated in seconds.

5 Discussion

The measured durations are close to the 83.8 fs transform limit, suggesting nearly transform-limited pulses within the analysis uncertainty. The autocorrelation fit uses the tutorial's two oscillatory terms and assumes a shared Gaussian envelope. Its fringe calibration converts oscilloscope time to optical delay under a locally constant scan-speed assumption. We exclude `vor2` because its shorter scan window truncates the envelope wings and its sampling interval

Table 1: Exploratory single-exponential fits to the TG and pump–probe traces. Periods and wave numbers use the unverified assumption that filename suffixes are periods in micrometres.

File	Assumed Λ (µm)	q (m^{-1})	τ (dataset units)
`GaAs_RT_PP_X.dat`	–	0	1.096
`GaAs_RT_5.6_X.dat`	5.6	1.122×10^6	2.554
`GaAs_RT_6.5_X.dat`	6.5	9.666×10^5	2.725
`GaAs_RT_8_X.dat`	8.0	7.854×10^5	2.960
`GaAs_RT_12_X.dat`	12.0	5.236×10^5	3.238

differs from the other before-compressor scans; the file is still displayed for reference in Fig. 8. Systematic effects from scan nonlinearity, model simplification, and instrument response are not included in the repeat-to-repeat standard deviations. The tutorial distinguishes carrier thermalization (roughly 100 fs) from subsequent energy cooling on a picosecond scale. A time-resolved Raman study of GaAs reported an average electron–LO-phonon scattering time of about 165 fs below an injected density of $10^{17}\,\mathrm{cm}^{-3}$; this microscopic scattering time is not interchangeable with the measured carrier cooling time, which depends on carrier distribution, density, and energy transfer to the lattice [2, 3]. The supplied traces do not identify a calibrated hot-carrier temperature or energy-relaxation observable, so neither a cooling constant nor an average electron–phonon coupling coefficient can be extracted.

The TG delay traces and exploratory single-exponential fits are shown in Fig. 6. Under the provisional period interpretation, the fitted rate versus q^2 regression has slope -2.98×10^{-13} in (dataset unit)$^{-1}\,\mathrm{m}^2$, intercept 0.647 (dataset unit)$^{-1}$, and $R^2 = 0.350$. The negative slope is incompatible with the simple diffusion model, which predicts a non-negative D_a, and the fitted TG lifetimes in Table 1 are longer than the $q = 0$ pump–probe lifetime. These inconsistencies show that the present fit/calibration does not recover the expected decomposition. Possible causes include an incorrect interpretation of suffixes or files, uncalibrated delay units, model inadequacy, and additional relaxation channels; the files do not let us select among them. Without verified grating periods and delay calibration, assigning a slope to D_a would introduce an unjustified scale factor. Therefore no numerical recombination time in seconds, diffusivity, or mobility is inferred. Literature shows that ambipolar diffusion can depend on carrier density; for example, calculations for GaAs include density-dependent many-body and degeneracy effects [4]. Comparison to reported values would require the sample's excitation density, temperature, calibrated grating periods, and physical time units. A repeat with those calibrations and instrument-response-convolved fits would permit a meaningful determination.

The principal uncertainty sources are local scan-speed variation in the autocorrelation conversion; finite fitting windows and the assumed Gaussian envelope; background and detector-response effects; spectral wavelength/intensity calibration; and unverified meanings and units of the TG files. The reported pulse-width scatter does not include these contributions. The data also omit pump fluence, sample temperature, carrier density, and a physical delay calibration, which are needed for quantitative comparisons across samples.

6 Conclusion

The spectrum gives an FWHM bandwidth of 10.92 nm (about 5.26 THz) and a Gaussian transform limit of 83.8 fs. Fits to the raw autocorrelation traces, excluding the truncated `vor2` scan, give (81.5 ± 0.3) fs before and (79.0 ± 0.1) fs after the compressor under the local constant-scan-speed assumption. The TG data still lack the calibrations needed for reliable cooling, recombination, diffusivity, and mobility measurements. The accompanying Python script performs the analysis and generates the plots.

References

[1] Institute of Physics II, University of Cologne, *M2.10: Ultrafast Transient Grating Spectroscopy*, practical-course tutorial, version December 2019, especially Eqs. (4.2)–(4.14) and Secs. 5–6.

[2] J. A. Kash, J. C. Tsang, and J. M. Hvam, "Subpicosecond Time-Resolved Raman Spectroscopy of LO Phonons in GaAs," *Physical Review Letters* **54**, 2151 (1985), doi:10.1103/PhysRevLett.54.2151.

[3] R. F. Leheny, J. Shah, R. L. Fork, C. V. Shank, and A. Migus, "Dynamics of hot carrier cooling in photo-excited GaAs," *Solid State Communications* **31**, 809–813 (1979), doi:10.1016/0038-1098(79)90393-4.

[4] J. F. Young and H. M. van Driel, "Ambipolar diffusion of high-density electrons and holes in Ge, Si, and GaAs: Many-body effects," *Physical Review B* **26**, 2147 (1982), doi:10.1103/PhysRevB.26.2147.

A Plots

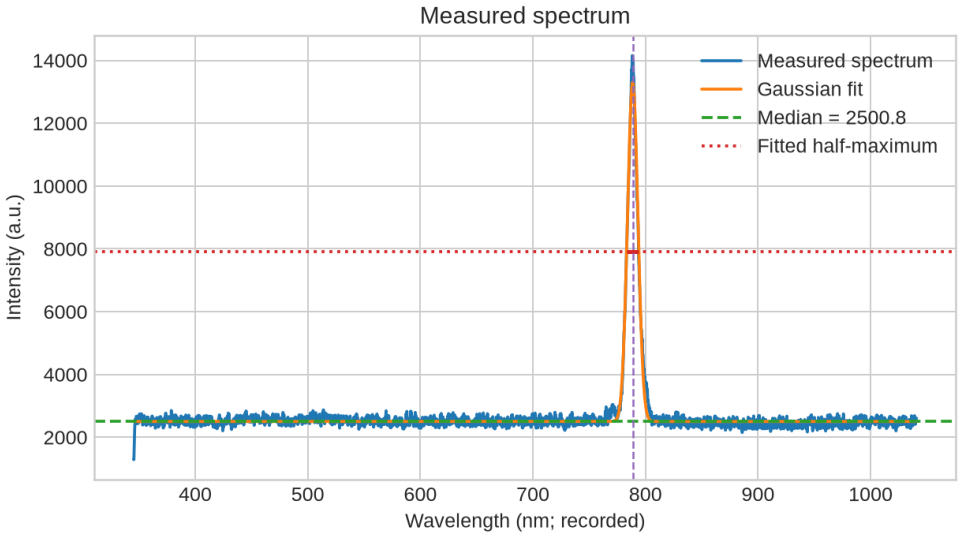


Figure 5: Measured spectrum, Gaussian fit, median level, and fitted half-maximum bandwidth.

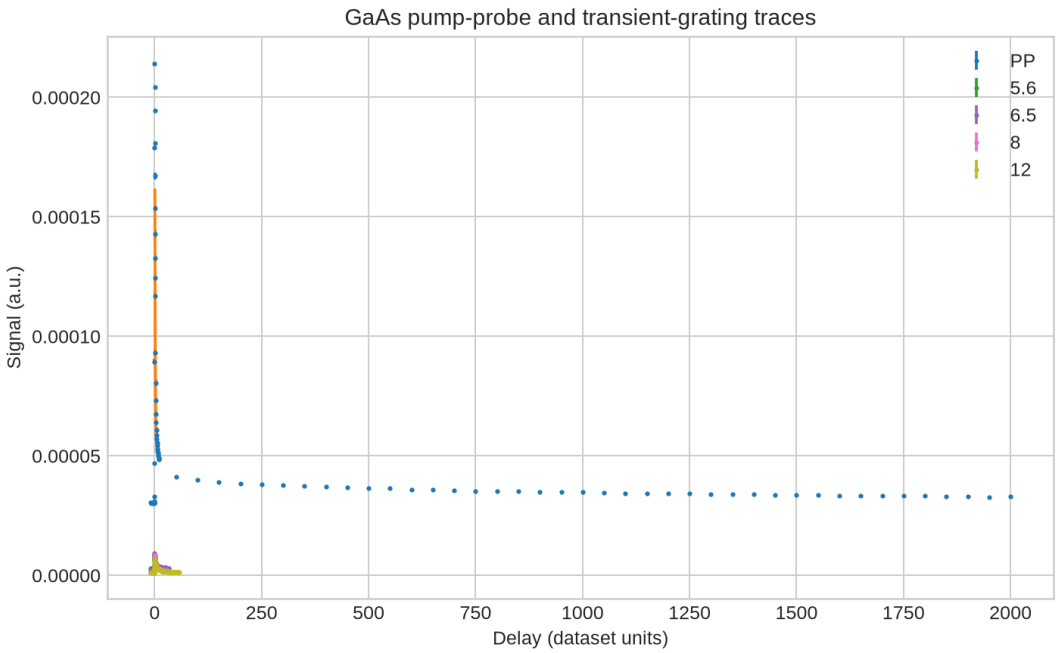


Figure 6: Mean delay traces and exploratory single-exponential fits. Repeated columns are averaged.

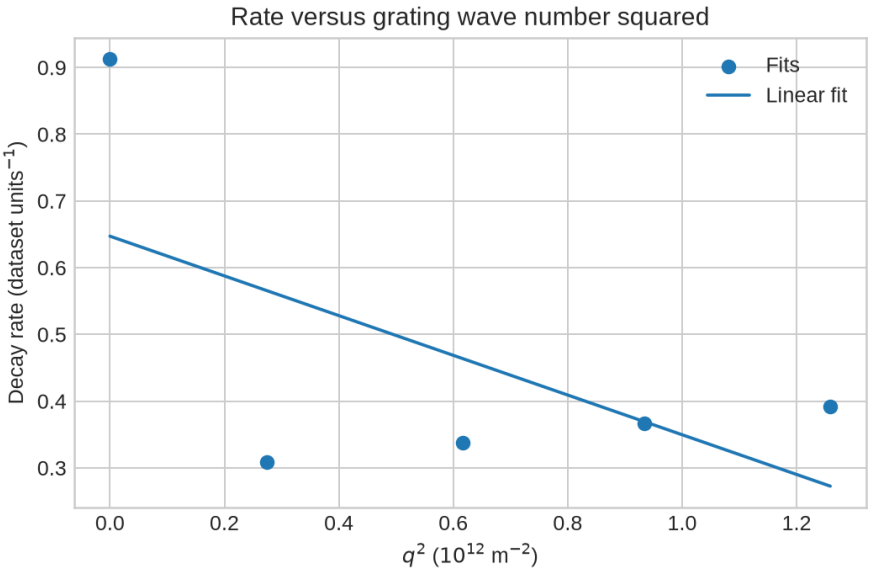


Figure 7: Exploratory fitted rate against nominal q^2. The poor linear consistency and missing period calibration prevent extracting a reliable diffusivity.

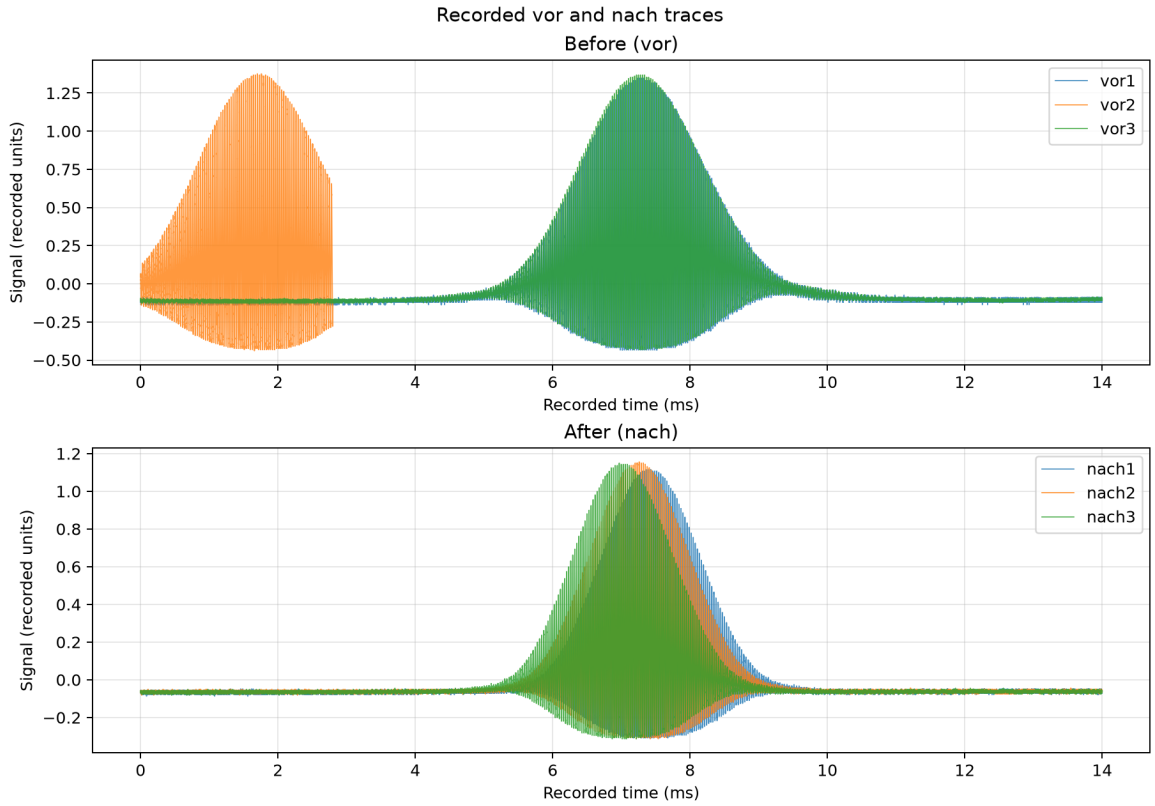


Figure 8: Recorded auxiliary traces grouped as before (vor) and after (nach). The horizontal axis uses recorded time converted to milliseconds.

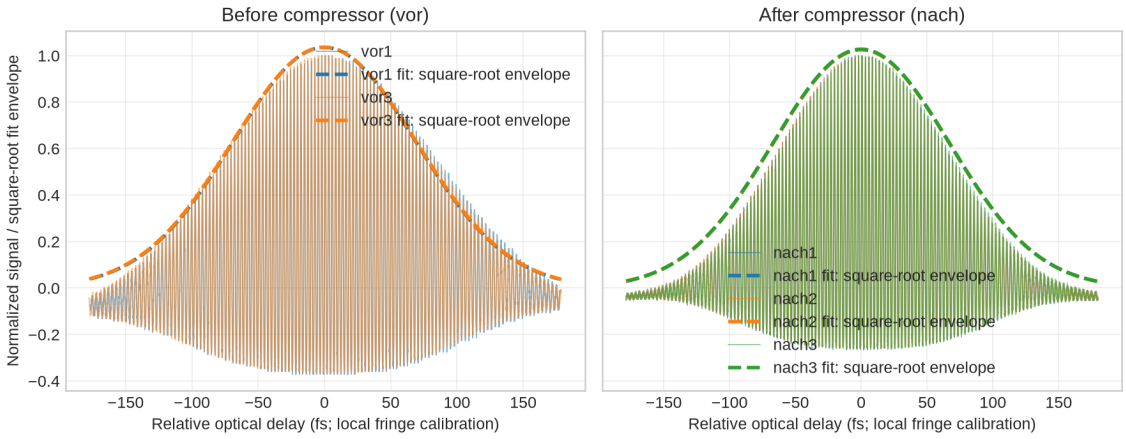


Figure 9: Raw autocorrelation traces after local fringe calibration, with the square root of each fitted Gaussian envelope overlaid. The fit uses the tutorial-based model with ω and 2ω terms. The truncated `vor2` trace is excluded from these fits.